

Multiple Choice

1. **Answer:** D

Options: A) Glomerular capillary pressure and blood volume; B) Diameter of the efferent arterioles and plasma protein concentration; C) Renal blood flow and the concentration of sodium in the distal tubule; D) Arterial blood pressure and diameter of the afferent arterioles; E) Kidney size and blood urea nitrogen levels

Note: Correct option: D - Arterial blood pressure and diameter of the afferent arterioles

2. **Answer:** A

Options: A) Oxygen is used to maintain the temperature of cells.; B) Oxygen is used by cells to produce alcohol.; C) Oxygen is used to produce carbon dioxide in cells.; D) Oxygen is solely responsible for stimulating brain activity.; E) Oxygen is used to synthesize proteins in cells.

Note: Correct option: A - Oxygen is used to maintain the temperature of cells.

3. **Answer:** E

Options: A) Epilepsy is the result of a chromosomal aneuploidy, similar to Down syndrome.; B) Epilepsy is always inherited in an X-linked recessive pattern.; C) Epilepsy is a dominant trait.; D) Epilepsy is caused by an autosomal dominant trait.; E) Epilepsy is due to an autosomal recessive trait.

Note: Correct option: E - Epilepsy is due to an autosomal recessive trait.

4. **Answer:** A

Options: A) that humans evolved from apes.; B) that evolution occurs.; C) that evolution occurs through a process of gradual change.; D) that the Earth is older than a few thousand years.; E) a mechanism for how evolution occurs.

Note: Correct option: A - that humans evolved from apes.

5. **Answer:** A

Options: A) Mimetic coloration; B) Aposomatic coloration; C) Deceptive markings; D) Batesian mimicry; E) Flash coloration

Note: Correct option: A - Mimetic coloration

True / False

6. **Answer:** False

Options: A) True; B) False

Note: This study showed that FDG is not a good surrogate tracer for tumor hypoxia under either ambient or hypoxic conditions. Only specific hypoxia tracers should be used to measure tumor hypoxia.

7. **Answer:** True

Options: A) True; B) False

Note: The prevalence of vagino-rectal colonization by Streptococcus group B in the pregnant women from Melilla is within the national estimated figures, however it is different if they are from Muslim or Christian culture, being higher in the Muslim population. On one hand both prevalences are within the national statistics, and on the other hand it is observed that there is not any difference according to age.

8. **Answer:** True

Options: A) True; B) False

Note: GFRUPs procedure was applicable in most cases. The main difficulties were anticipating the correct date for the meeting and involving nurses in the procedure. Children for whom the procedure was interrupted because of clinical improvement and who survived in poor condition without a formal decision pointed out the need for medical criteria for questioning, which should systematically lead to a formal decision-making process.

9. **Answer:** True

Options: A) True; B) False

Note: According to the results from this meta-analysis, the risk of grade 3 and 4 CVAEs in patients who were receiving AIs was higher compared with the risk in patients who were receiving tamoxifen, and the difference reached statistical significance. However, the AD was relatively low, and from 160 to 180 patients had to be treated to produce 1 event.

10. **Answer:** True

Options: A) True; B) False

Note: The current literature suggests that dexamethasone can be used as an effective alternative to prednisone in the treatment of mild to moderate acute asthma exacerbations in children, with the added benefits of improved compliance, palatability, and cost. However, more research is needed to examine the role of dexamethasone in hospitalized children.

Long Form Response

11. **Answer:** Cause tremors, spasms, or even death. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The baby contracted it from a contaminated surface.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key